

Supplementary Table S1. Complete Feature Dictionary and Prediction-Time Availability Audit

Decision-Aware Hierarchical Multi-Model Orchestration Framework for Explainable and Consistent Multi-Output Traffic Accident Severity Prediction

Scope and terminology. The intended prediction setting is post-crash police-record-based assessment. The 278-column processed analytical table described in the manuscript contains contextual fields together with outcome fields retained only for target construction. Thirteen outcome/target-equivalent fields were excluded from model inputs, leaving 265 contextual predictors; three dynamically generated physical-severity features were then added, yielding the 268-column audited final predictor pool. The total-injury branch additionally uses five branch-specific derived variables listed in Panel B. Raw identifiers, direct coordinates, road name, and raw date/time fields listed in Panel C were excluded before/at model construction and are not predictors. The Turkish source field Yoya_Kazazede is reported here using the corrected English label “Road-User Type.” Its one-hot levels identify the role of an involved road user (Passenger or Pedestrian) and do not encode whether that individual was injured or killed and do not contain an injury/fatality count.

Leakage-audit principle. A variable is treated as admissible when it is available in the defined post-crash police-record stage and does not directly reveal the modeled outcome. Direct injury/fatality outcomes, target-equivalent derivatives, and their counts are excluded. “Not specified” or “not detected” one-hot levels in retained contextual fields are source-record categories; they are not missingness indicators generated from missing target values. Training-derived encodings are fitted using the training partition only and are transferred unchanged to validation/test data.

Panel A. Retained final predictor dictionary (268 variables)

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
1	YolunYasalHizLimiti	Legal road speed limit	Roadway / location; Numeric/count	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
2	YolSeritSayisi	Number of road lanes	Roadway / location; Numeric/count	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
3	YolSeritGenisligi	Lane width	Roadway / location; Numeric/count	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
4	KazaKm	Crash location kilometer marker	Roadway / location; Numeric/count	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
5	KazaMetre	Crash location meter offset	Roadway / location; Numeric/count	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
6	YolKilometre	Road kilometer reference	Roadway / location; Numeric/count	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
7	YolMetre	Road meter reference	Roadway / location; Numeric/count	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
8	Kazali_1	Crash province: Adana	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
9	Kazali_7	Crash province: Antalya	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
10	Kazali_33	Crash province: Mersin	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
11	BirimGrubu_Bölge	Police/unit area group: Regional unit	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
12	BirimGrubu_Şehir İçi	Police/unit area group: Urban unit	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
13	KazaYerYeriDışı_1-Yerl.Yeri	Settlement context: Within settlement	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
14	KazaYerYeriDışı_2-Yerl.Yeri Disi	Settlement context: Outside settlement	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
15	YolunTipi_1-Bölünmüş Yol	Road type: Divided road	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
16	YolunTipi_2-Tek Yönlü Yol	Road type: One-way road	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
17	YolunTipi_3-İki Yönlü Yol	Road type: Two-way road	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
18	YolunTipi_4-Diger	Road type: Other	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
19	YolunKaplamaşı_1-Asfalt	Road pavement material: Asphalt	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
20	YolunKaplamaşı_3-Beton	Road pavement material: Concrete	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
21	YolunKaplamaşı_4-Parke	Road pavement material: Paving blocks	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
22	YolunKaplamaşı_Diger	Road pavement material: Other	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
23	YolunSınıfı_1-Cadde	Road class: Avenue/main street	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
24	YolunSınıfı_2-Sokak	Road class: Street	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
25	YolunSınıfı_3-Otoyol	Road class: Motorway	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
26	GeoYatay_1-Düz yol	Horizontal road alignment: Straight road	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
27	GeoYatay_2-Viraj	Horizontal road alignment: Curve	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
28	GeoDüşey_1-Eğimsiz	Vertical road alignment: Level	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
29	GeoDüşey_2-Eğimli	Vertical road alignment: Grade/slope	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
30	GeoKavşak_1-Üç Yönlü (T)	Intersection geometry: Three-way (T) intersection	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
31	GeoKavşak_2-Üç Yönlü (Y)	Intersection geometry: Three-way (Y) intersection	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

Supplementary Table S1 - Feature Dictionary

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32	GeoKavşak_3-Dört Yönlü	Intersection geometry: Four-way intersection	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
33	GeoKavşak_4-Dönel Kavsak	Intersection geometry: Roundabout	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
34	GeoKavşak_6-Diger Kavsak Çesidi	Intersection geometry: Other intersection type	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
35	GeoKavşak_8-Kavşak Yok	Intersection geometry: No intersection	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
36	GeoGeçit_3-Okul Geçiti	Crossing type: School crossing	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
37	GeoGeçit_4-Yaya Geçiti	Crossing type: Pedestrian crossing	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
38	GeoGeçit_5-Geçit Yok	Crossing type: No crossing	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
39	GeoGeçit_Demiryolu Geçidi	Crossing type: Railway crossing	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
40	Gündurumu_1-Gündüz	Light/day condition: Daylight	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
41	Gündurumu_2-Gece	Light/day condition: Night	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
42	Gündurumu_3-AlacaKaranlık	Light/day condition: Twilight	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
43	Havadurumu_1-Açık	Weather condition: Clear	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
44	Havadurumu_10-Bulutlu	Weather condition: Cloudy	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
45	Havadurumu_2-Sis/Duman	Weather condition: Fog/smoke	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
46	Havadurumu_3-Yagmur	Weather condition: Rain	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
47	Havadurumu_4-Kar	Weather condition: Snow	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
48	Havadurumu_8-Kuvvetli Rüzgar	Weather condition: Strong wind	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
49	YolunYüzeyi_1-Kuru	Road surface condition: Dry	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
50	YolunYüzeyi_2-Islak/Nemli	Road surface condition: Wet/damp	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

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51	YolunYüzeyi_6-Diger Kaygan Yüzey	Road surface condition: Other slippery surface	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
52	YolunYüzeyi_Karli	Road surface condition: Snow-covered	Environment; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
53	KazaikYardım_1-Saglik Ekibi	First aid / first responder: Medical team	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
54	KazaikYardım_2-Trafik Zabitası	First aid / first responder: Traffic police	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
55	KazaikYardım_3-Vatandas	First aid / first responder: Citizen/by-stander	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
56	KazaOlusTuru_1-Karsilikli Çarpisma	Crash type: Head-on collision	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
57	KazaOlusTuru_11-Devrilme/Savrulma/Takla	Crash type: Rollover/skid/overturn	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
58	KazaOlusTuru_12-Yoldan Çıkma	Crash type: Run-off-road	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
59	KazaOlusTuru_15-Park Etmiş Araca Çarpma	Crash type: Collision with parked vehicle	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
60	KazaOlusTuru_2-Arkadan Çarpma	Crash type: Rear-end collision	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
61	KazaOlusTuru_3-Yandan Çarpma	Crash type: Side-impact collision	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
62	KazaOlusTuru_4-Yan Yana Çarpisma	Crash type: Sideswipe collision	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
63	KazaOlusTuru_5-Duran Araca Çarpma	Crash type: Collision with stopped vehicle	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
64	KazaOlusTuru_6-Zincirleme Çarpisma	Crash type: Chain collision	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
65	KazaOlusTuru_7-Çoklu Çarpisma	Crash type: Multiple collision	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
66	KazaOlusTuru_8-Engel/Cisim Ile Çarpisma	Crash type: Collision with obstacle/object	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
67	KazaOlusTuru_9-Yayaya Çarpma	Crash type: Collision with pedestrian	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
68	KazaAracSayısı_1-Tek Araçlı	Vehicle-count category: Single-vehicle	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
69	KazaAracSayısı_2-İki Araçlı	Vehicle-count category: Two-vehicle	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

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70	KazaAracSayisi_3-Çok Araçlı	Vehicle-count category: Multi-vehicle	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
71	İlkcarpismaYeri_Banket Üzerinde	First collision location: On shoulder	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
72	İlkcarpismaYeri_Diger	First collision location: Other	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
73	İlkcarpismaYeri_Orta Refüjde	First collision location: On median	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
74	İlkcarpismaYeri_Yaya/Bisiklet Yolu	First collision location: Pedestrian/bicycle path	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
75	İlkcarpismaYeri_Yol Kenarında (Banket Disi)	First collision location: Roadside (outside shoulder)	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
76	İlkcarpismaYeri_Yol Üzerinde	First collision location: On roadway	Crash configuration / scene; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
77	YolGenisligi_10 m ve üzeri	Road width category: 10 m or wider	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
78	YolGenisligi_300 cm	Road width category: 300 cm	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
79	YolGenisligi_400 cm	Road width category: 400 cm	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
80	YolGenisligi_500 cm	Road width category: 500 cm	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
81	YolGenisligi_600 cm	Road width category: 600 cm	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
82	YolGenisligi_700 cm	Road width category: 700 cm	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
83	YolGenisligi_800 cm	Road width category: 800 cm	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
84	YolGenisligi_900 cm	Road width category: 900 cm	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
85	EmnŞeritiBanket_Var	Emergency lane/shoulder: Present	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
86	EmnŞeritiBanket_Yok	Emergency lane/shoulder: Absent	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
87	Yolşeritçizgisi_1-Var	Lane markings: Present	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
88	Yolşeritçizgisi_Yok	Lane markings: Absent	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

Supplementary Table S1 - Feature Dictionary

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89	TrafikİşaretLevhası_1-Var	Traffic signs: Present	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
90	TrafikİşaretLevhası_Yok	Traffic signs: Absent	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
91	YayageçitÖzelligi_1-Üst Geçit	Pedestrian crossing feature: Overpass	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
92	YayageçitÖzelligi_2-Alt Geçit	Pedestrian crossing feature: Underpass	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
93	YayageçitÖzelligi_3-Sinyalize Geçit	Pedestrian crossing feature: Signalized crossing	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
94	YayageçitÖzelligi_4-Sinyalizasyonsuz Geçit	Pedestrian crossing feature: Unsignalized crossing	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
95	YayageçitÖzelligi_5-Diğer	Pedestrian crossing feature: Other	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
96	YayageçitÖzelligi_6-Geçit Yok	Pedestrian crossing feature: No crossing	Roadway / location; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
97	HaftaGün_CUMA	Day of week: Friday	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
98	HaftaGün_CUMARTESİ	Day of week: Saturday	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
99	HaftaGün_PAZAR	Day of week: Sunday	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
100	HaftaGün_PAZARTESİ	Day of week: Monday	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
101	HaftaGün_PERSEMBE	Day of week: Thursday	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
102	HaftaGün_SALI	Day of week: Tuesday	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
103	HaftaGün_ÇARŞAMBA	Day of week: Wednesday	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
104	KazaSaatDilimi_Encoded_0	Time-of-day bin: encoded level 0	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
105	KazaSaatDilimi_Encoded_1	Time-of-day bin: encoded level 1	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
106	KazaSaatDilimi_Encoded_2	Time-of-day bin: encoded level 2	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
107	KazaSaatDilimi_Encoded_3	Time-of-day bin: encoded level 3	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

Supplementary Table S1 - Feature Dictionary

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
108	KazaSaatDilimi_Encoded_4	Time-of-day bin: encoded level 4	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
109	KazaSaatDilimi_Encoded_5	Time-of-day bin: encoded level 5	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
110	KazaSaatDilimi_Encoded_6	Time-of-day bin: encoded level 6	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
111	KazaSaatDilimi_Encoded_7	Time-of-day bin: encoded level 7	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
112	KazaSaatDilimi_Encoded_8	Time-of-day bin: encoded level 8	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
113	KazaSaatDilimi_Encoded_9	Time-of-day bin: encoded level 9	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
114	KazaSaatDilimi_Encoded_10	Time-of-day bin: encoded level 10	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
115	KazaSaatDilimi_Encoded_11	Time-of-day bin: encoded level 11	Temporal (derived/encoded); One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
116	kazayaKarisAnAracSayisi	Number of vehicles involved in the crash	Crash configuration / scene; Numeric/count	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
117	AracCinsi_1-Bisiklet	Vehicle type: Bicycle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
118	AracCinsi_10-Otobüs	Vehicle type: Bus	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
119	AracCinsi_13-Özel Amaçlı	Vehicle type: Special-purpose vehicle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
120	AracCinsi_3-Motorlu Bisiklet	Vehicle type: Moped	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
121	AracCinsi_4-Motosiklet	Vehicle type: Motorcycle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
122	AracCinsi_5-Otomobil	Vehicle type: Passenger car	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
123	AracCinsi_6-Minibüs	Vehicle type: Minibus	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
124	AracCinsi_7-Kamyonet	Vehicle type: Light truck/van	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
125	AracCinsi_8-Kamyon	Vehicle type: Truck	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
126	AracCinsi_9-Çekici	Vehicle type: Tractor unit	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

Supplementary Table S1 - Feature Dictionary

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
127	AracKazaSonrasıKonumu_1-Kaza Yerinde	Vehicle status/location after crash: At crash scene	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
128	AracKazaSonrasıKonumu_2-Çarpıp Kaçma	Vehicle status/location after crash: Hit-and-run	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
129	AracKazaSonrasıKonumu_3-Kaza Yerinden Zorunluluk Halinde Uzaklaşma	Vehicle status/location after crash: Left crash scene due to necessity	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
130	AracKullanımamacı_1-Özel	Vehicle use purpose: Private	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
131	AracKullanımamacı_2-Ticari	Vehicle use purpose: Commercial	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
132	AracKullanımamacı_3-Emniyet	Vehicle use purpose: Police/security	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
133	AracKullanımamacı_4-Askeri	Vehicle use purpose: Military	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
134	AracKullanımamacı_5-Diger Kamu KuruluS	Vehicle use purpose: Other public institution	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
135	AracKullanımamacı_6-Zirai	Vehicle use purpose: Agricultural	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
136	AracKullanımamacı_7-Yabancı	Vehicle use purpose: Foreign	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
137	AracKazaAnıKullanımSekli_11 Ş.İçi Minibüs (Dolmuş)	Vehicle use at time of crash: Urban minibus (dolmuş)	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
138	AracKazaAnıKullanımSekli_12 Şehirlerar. Minibüs	Vehicle use at time of crash: Intercity minibus	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
139	AracKazaAnıKullanımSekli_14 Ticari Taksı	Vehicle use at time of crash: Commercial taxi	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
140	AracKazaAnıKullanımSekli_16 Diğer Otomobil	Vehicle use at time of crash: Other passenger car	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
141	AracKazaAnıKullanımSekli_22 Yük Taş. Diğer Araç	Vehicle use at time of crash: Other goods-transport vehicle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
142	AracKazaAnıKullanımSekli_23 Yolcu Taş. Diğer Araç	Vehicle use at time of crash: Other passenger-transport vehicle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
143	AracKazaAnıKullanımSekli_28 Yabancı Araç	Vehicle use at time of crash: Foreign-registered vehicle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
144	AracKazaAnıKullanımSekli_31 Emniyet Aracı	Vehicle use at time of crash: Police/security vehicle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
145	AracKazaAnıKullanımSekli_37 Bisiklet	Vehicle use at time of crash: Bicycle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

Supplementary Table S1 - Feature Dictionary

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
146	AracKazaAniKullanımSekli_38 Elektrikli Bisiklet	Vehicle use at time of crash: Electric bicycle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
147	AracKazaAniKullanımSekli_45 Diğer Arac	Vehicle use at time of crash: Other vehicle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
148	AracKazaAniKullanımSekli_Servis	Vehicle use at time of crash: Service vehicle	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
149	AracHareketi_1-Dogru istikamette gidiyor	Vehicle movement: Proceeding straight	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
150	AracHareketi_10 U-Dönüşü yapıyor	Vehicle movement: Making a U-turn	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
151	AracHareketi_11-Park halinde	Vehicle movement: Parked	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
152	AracHareketi_14-Tek yön.böl.yol.ter yön.gid	Vehicle movement: Traveling against traffic on one-way/divided road	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
153	AracHareketi_2-Serit degistiriyor	Vehicle movement: Changing lane	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
154	AracHareketi_3-Öndeki aracın solundan	Vehicle movement: Passing/overtaking on left	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
155	AracHareketi_4-Öndeki aracın sağından	Vehicle movement: Passing on right	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
156	AracHareketi_5-Geriye dogru gidiyor	Vehicle movement: Reversing	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
157	AracHareketi_6-Trafik Şeridine katılıyor	Vehicle movement: Entering traffic lane	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
158	AracHareketi_7-Trafik Şeridinden ayrılıyor	Vehicle movement: Leaving traffic lane	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
159	AracHareketi_8-Sola dönüyor	Vehicle movement: Turning left	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
160	AracHareketi_9-Saga dönüyor	Vehicle movement: Turning right	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
161	AracHareketi_Yavasliyor/Duruyor	Vehicle movement: Slowing/stopping	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
162	AracDarbeBölümü_1-Ön	Vehicle impact area: Front	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
163	AracDarbeBölümü_10-Sag Ön Çamurluk	Vehicle impact area: Right front fender	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
164	AracDarbeBölümü_11-Sol Arka Çamurluk	Vehicle impact area: Left rear fender	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

Supplementary Table S1 - Feature Dictionary

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
165	AracDarbeBölümü_12-Sag Arka Çamurluk	Vehicle impact area: Right rear fender	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
166	AracDarbeBölümü_13-Aracin Üstü	Vehicle impact area: Vehicle roof/top	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
167	AracDarbeBölümü_14-Aracin Altı	Vehicle impact area: Vehicle underside	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
168	AracDarbeBölümü_2-Arka	Vehicle impact area: Rear	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
169	AracDarbeBölümü_3-Sol	Vehicle impact area: Left side	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
170	AracDarbeBölümü_4-Sag	Vehicle impact area: Right side	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
171	AracDarbeBölümü_5-Sol Ön Far	Vehicle impact area: Left front headlamp	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
172	AracDarbeBölümü_6-Sag Ön Far	Vehicle impact area: Right front headlamp	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
173	AracDarbeBölümü_7-Sol Arka Far	Vehicle impact area: Left rear lamp	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
174	AracDarbeBölümü_8-Sag Arka Far	Vehicle impact area: Right rear lamp	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
175	AracDarbeBölümü_9-Sol Ön Çamurluk	Vehicle impact area: Left front fender	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
176	AracHasarDerecesi_2-Hafif Hasar	Vehicle damage severity: Minor damage	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
177	AracHasarDerecesi_3-Fonksiyonel Hasar	Vehicle damage severity: Functional damage	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
178	AracHasarDerecesi_4-Hareket Edemez	Vehicle damage severity: Immobile/non-drivable	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
179	AracHasarDerecesi_5-Agir Hasar	Vehicle damage severity: Severe damage	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
180	AracHasarDerecesi_Hasarsiz	Vehicle damage severity: No damage	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
181	AracYakitTürü_1-Benzin	Vehicle fuel type: Gasoline	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
182	AracYakitTürü_12-Diger	Vehicle fuel type: Other	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
183	AracYakitTürü_2-Dizel	Vehicle fuel type: Diesel	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

Supplementary Table S1 - Feature Dictionary

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
184	AracYakıtTürü_5-Elektrik	Vehicle fuel type: Electric	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
185	AracYakıtTürü_Benzin/Dizel+LPG	Vehicle fuel type: Gasoline/diesel + LPG	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
186	AracYakıtTürü_CNG	Vehicle fuel type: CNG	Vehicle characteristics / dynamics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
187	Sürücü_Cinsiyet_1-Erkek	Driver sex: Male	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
188	Sürücü_Cinsiyet_2-Kadin	Driver sex: Female	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
189	Sürücü_ÖğrenimDurumu_1	Driver education status: source code 1	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
190	Sürücü_ÖğrenimDurumu_3	Driver education status: source code 3	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
191	Sürücü_ÖğrenimDurumu_4	Driver education status: source code 4	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
192	Sürücü_ÖğrenimDurumu_5	Driver education status: source code 5	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
193	Sürücü_AlkolSonucu_Alkolsüz	Driver alcohol result: No alcohol detected	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
194	Sürücü_AlkolSonucu_Az Alkollü	Driver alcohol result: Low alcohol level	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
195	Sürücü_AlkolSonucu_Dr-Alkollü	Driver alcohol result: Alcohol positive (source-coded category)	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
196	Sürücü_AlkolSonucu_Çok Alkollü	Driver alcohol result: High alcohol level	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
197	Sürücü_Kusuru1_Alkollü olarak araç kullanmak	Primary driver fault: Driving under the influence of alcohol	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
198	Sürücü_Kusuru1_Araç hızını yol, hava ve trafiğin gerektirdiği şartlara uydurmak	Primary driver fault: Failure to adapt vehicle speed to road, weather, and traffic conditions	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
199	Sürücü_Kusuru1_Arkadan çarpmak	Primary driver fault: Rear-end collision fault	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
200	Sürücü_Kusuru1_Aşırı hızla araç kullanmak	Primary driver fault: Driving at excessive speed	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
201	Sürücü_Kusuru1_Bisiklet, M.bisiklet ve Motosikletleri kurallara uymadan sürmek	Primary driver fault: Operating a bicycle/moped/motorcycle contrary to traffic rules	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
202	Sürücü_Kusuru1_Doğrultu değiştirme (dönüş) kurallarına uymamak	Primary driver fault: Failure to comply with turning/change-of-direction rules	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

Supplementary Table S1 - Feature Dictionary

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
203	Sürücü_Kusuru1_Eksik, bozuk veya uygun olmayan araç donanımıyla araç kullanmak	Primary driver fault: Driving with missing, defective, or unsuitable vehicle equipment	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
204	Sürücü_Kusuru1_Geçme yasağı olan yerlerden geçmek	Primary driver fault: Overtaking where prohibited	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
205	Sürücü_Kusuru1_Hatalı şekilde veya yasak olan yerlere park etmek	Primary driver fault: Improper or prohibited parking	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
206	Sürücü_Kusuru1_Kavşak, geçiş önceliğine uymamak	Primary driver fault: Failure to comply with intersection/right-of-way priority	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
207	Sürücü_Kusuru1_Kaza mahallinde durmamak, gerekli tedbirleri almamak ve yetkililere bildirmemek	Primary driver fault: Failure to stop, take required measures, or notify authorities after a crash	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
208	Sürücü_Kusuru1_Kurallara uygun olarak park etmiş araçlara çarpmak	Primary driver fault: Collision with a properly parked vehicle	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
209	Sürücü_Kusuru1_Kusursuz	Primary driver fault: No recorded fault	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
210	Sürücü_Kusuru1_Kırmızı ışık veya görevlinin dur işaretine uymamak	Primary driver fault: Failure to obey a red light or authorized stop signal	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
211	Sürücü_Kusuru1_Manevraları düzenleyen genel şartlara uymamak	Primary driver fault: Failure to comply with general maneuvering rules	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
212	Sürücü_Kusuru1-Taşıt giremez trafik işareti bulunan yerlere girmek	Primary driver fault: Entering an area marked as prohibited to vehicles	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
213	Sürücü_Kusuru1_Trafik güvenliği ile ilgili diğer kurallara uymamak	Primary driver fault: Failure to comply with other traffic-safety rules	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
214	Sürücü_Kusuru1_Yaya ve okul geçitlerinde yavaşlamamak, yayaalara geçiş hakkı vermemek	Primary driver fault: Failure to slow/give way at pedestrian or school crossings	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
215	Sürücü_Kusuru1_Şerit izleme ve değiştirme kuralı uymamak	Primary driver fault: Failure to comply with lane-keeping/lane-change rules	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
216	Yaş Aralığı_Ehliyetsiz	Driver age/license group: Unlicensed	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
217	Yaş Aralığı_Genç Sürücüler	Driver age/license group: Young drivers	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
218	Yaş Aralığı_Yaşlı Sürücüler	Driver age/license group: Older drivers	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
219	Yaş Aralığı_Çalışma Çağı	Driver age/license group: Working-age drivers	Driver characteristics / behavior; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
220	Yoya_Kazazede_1-Yolcu	Road-user type: Passenger	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Role indicator only (Road-User Type); does not encode injury/death status or casualty count.
221	Yoya_Kazazede_2-Yaya	Road-user type: Pedestrian	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Role indicator only (Road-User Type); does not encode injury/death status or casualty count.

Supplementary Table S1 - Feature Dictionary

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
222	Yoya_Cinsiyet_1-Erkek	Passenger/pedestrian sex: Male	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
223	Yoya_Cinsiyet_2-Kadin	Passenger/pedestrian sex: Female	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
224	Yaya_Kusuru1_Geçit ve kavşakların bulunmadığı yerlerde geçme kurallarına uymamak	Primary pedestrian fault: Failure to comply with crossing rules away from designated crossings/intersections	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
225	Yaya_Kusuru1_Karşıdan karşıya geçişlerde trafik kurallarına uymamak	Primary pedestrian fault: Failure to comply with traffic rules while crossing	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
226	Yaya_Kusuru1_Kusursuz	Primary pedestrian fault: No recorded fault	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
227	Yaya_Kusuru1-Taşıt yolu üzerinde trafiği tehlikeye düşürücü hareketlerde bulunmak	Primary pedestrian fault: Unsafe behavior on the roadway that endangers traffic	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
228	Yaya_Kusuru1-Taşıt yoluna girmek	Primary pedestrian fault: Entering the roadway	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
229	Yaya_Kusuru1-Taşıt yolunda sol kenardan gitmemek	Primary pedestrian fault: Failure to walk on the prescribed left side of the roadway	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
230	Yaya_Kusuru1-Taşıt yolunda yürümek	Primary pedestrian fault: Walking on the roadway	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
231	Yaya_Kusuru1_Trafik güvenliği ile ilgili diğer kurallara uymamak	Primary pedestrian fault: Failure to comply with other traffic-safety rules	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
232	Yaya_Kusuru1_Trafik ışık ve işaretlerine uymamak	Primary pedestrian fault: Failure to obey traffic lights/signs	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
233	Yolcu_Kusuru1_Açık Yük Üzerinde Gayri Nizami ve Tehlikeli Şekilde Seyahat Etmek_x000D_	Primary passenger fault: Traveling unsafely/irregularly on an open load	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
234	Yolcu_Kusuru1_Emniyet kemeri takmamak	Primary passenger fault: Not wearing a seat belt	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
235	Yolcu_Kusuru1_Gerekli tedbirleri almadan taşıttan inmek	Primary passenger fault: Exiting the vehicle without required precautions	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
236	Yolcu_Kusuru1_Kask kullanmamak	Primary passenger fault: Not wearing a helmet	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
237	Yolcu_Kusuru1_Kusursuz	Primary passenger fault: No recorded fault	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
238	Yolcu_Kusuru1-Taşıt Dışında (Çamurluk v.s) Gayri Nizami Seyahat Etmek_x000D_	Primary passenger fault: Traveling unsafely outside the passenger compartment (e.g., fender)	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
239	Yolcu_Kusuru1_Trafik güvenliği ile ilgili diğer kurallara uymamak	Primary passenger fault: Failure to comply with other traffic-safety rules	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
240	Yolcu_KemerKask_0-Belirtilmemis	Passenger restraint/helmet use: Not specified	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Source-record category; not generated from missing injury/fatality target values.

Supplementary Table S1 - Feature Dictionary

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
241	Yolcu_KemerKask_1-Emniyet Kemer Takili	Passenger restraint/helmet use: Seat belt worn	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
242	Yolcu_KemerKask_2-Emniyet Kemer Takili Degil	Passenger restraint/helmet use: Seat belt not worn	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
243	Yolcu_KemerKask_3-Emniyet Kemer Zorunlu Degil	Passenger restraint/helmet use: Seat belt not required	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
244	Yolcu_KemerKask_4-Kask Takili	Passenger restraint/helmet use: Helmet worn	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
245	Yolcu_KemerKask_6-Kaski Yok	Passenger restraint/helmet use: No helmet	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
246	Yaya_Hareketi_1-Karsiya Geçiyor	Pedestrian movement: Crossing roadway	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
247	Yaya_Hareketi_2-Duruyor	Pedestrian movement: Standing/stationary	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
248	Yaya_Hareketi_3-Yolda Yürüyor/Kosuyor	Pedestrian movement: Walking/running on road	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
249	Yaya_Hareketi_4-Kaldır.Yol.dış.Dur/Yür	Pedestrian movement: Standing/walking outside sidewalk/roadway (source-coded)	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
250	Yaya_Hareketi_4-Tespit Edilemedi	Pedestrian movement: Not detected/undetermined	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Source-record category; not generated from missing injury/fatality target values.
251	Yolcu_Yeri_0-Belirtilmemis	Passenger position: Not specified	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Source-record category; not generated from missing injury/fatality target values.
252	Yolcu_Yeri_1-Ön Koltuk	Passenger position: Front seat	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
253	Yolcu_Yeri_2-AraKoridor	Passenger position: Middle/aisle	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
254	Yolcu_Yeri_3-Arka Koltuk	Passenger position: Rear seat	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
255	Yolcu_Yeri_4-Kasa İçi	Passenger position: Cargo bed	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
256	Yolcu_Yeri_5-Araç Üstü	Passenger position: On vehicle exterior/top	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
257	Yaş Aralığı_Ergen	Passenger/pedestrian age group: Adolescent	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
258	Yaş Aralığı_Genç	Passenger/pedestrian age group: Young	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
259	Yaş Aralığı_Yaşlı	Passenger/pedestrian age group: Older adult	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.

Supplementary Table S1 - Feature Dictionary

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
260	Yaş Aralığı_Yetişkin	Passenger/pedestrian age group: Adult	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
261	YayaAlkolSonucu_Alkolsüz	Pedestrian alcohol result: No alcohol detected	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
262	YayaAlkolSonucu_Az Alkollü	Pedestrian alcohol result: Low alcohol level	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
263	YayaAlkolSonucu_Belirtilmemiş	Pedestrian alcohol result: Not specified	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Source-record category; not generated from missing injury/fatality target values.
264	YayaAlkolSonucu_Dr-Alkollü	Pedestrian alcohol result: Alcohol positive (source-coded category)	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
265	YayaAlkolSonucu_Çok Alkollü	Pedestrian alcohol result: High alcohol level	Involved road-user characteristics; One-hot/binary indicator	Yes - available in post-crash police record	Retained predictor. Post-crash contextual/exposure predictor; does not directly encode a modeled injury/fatality outcome.
266	Kritik_Enerji	Collision Energy (engineered): number of involved vehicles × legal speed limit squared	Engineered physical-severity feature; Derived numeric	Yes - derived from retained post-crash predictors	Retained predictor. Deterministically derived from retained post-crash contextual predictors; no target value used.
267	Motosiklet_Hiz_Riski	Motorcycle Speed Risk (engineered): motorcycle indicator × legal speed limit	Engineered physical-severity feature; Derived numeric	Yes - derived from retained post-crash predictors	Retained predictor. Deterministically derived from retained post-crash contextual predictors; no target value used.
268	Hasar_Siddeti	Damage Severity (engineered): severe-damage indicator + immobility indicator	Engineered physical-severity feature; Derived numeric	Yes - derived from retained post-crash predictors	Retained predictor. Deterministically derived from retained post-crash contextual predictors; no target value used.

Panel B. Total-injury branch-specific derived variables

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
B1	Kapasite_Skoru	Vehicle capacity/exposure proxy derived from vehicle-type/use indicators	Total-injury branch-specific derived feature; Derived numeric	Available by derivation from post-crash vehicle descriptors	Retained branch-specific predictor. Deterministic exposure proxy; does not use an observed injury/fatality count.
B2	Risk_Indeksi	Risk Index: fixed ordinal score based on road type and heavy-vehicle involvement	Total-injury branch-specific derived feature; Derived numeric	Available by derivation from post-crash road/vehicle descriptors	Retained branch-specific predictor. Fixed domain-informed encoding; not a target value or calibrated outcome probability.
B3	Arac_Risk_Skoru	Vehicle Risk Score: training-set historical injury-average encoding by vehicle type	Total-injury branch-specific training-derived feature; Training-derived numeric	Derived from training partition only; mapping transferred unchanged to validation/test	Retained branch-specific predictor. Training-only target encoding; validation/test outcomes are not used to compute the mapping.
B4	Yol_Risk_Skoru	Road Risk Score: training-set historical injury-average encoding by road type	Total-injury branch-specific training-derived feature; Training-derived numeric	Derived from training partition only; mapping transferred unchanged to validation/test	Retained branch-specific predictor. Training-only target encoding; validation/test outcomes are not used to compute the mapping.
B5	Kaza_Cluster_ID	Accident Cluster ID from K-means clustering of retained predictor variables	Total-injury branch-specific unsupervised feature; Derived categorical/numeric ID	Cluster model fit without validation/test target information	Retained branch-specific predictor. Unsupervised structural descriptor; no injury/fatality target is used to form cluster assignments.

Panel C. Fields excluded from predictor matrices

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
C1	Kazayılı_x	Crash year (used for temporal partitioning, not as model predictor)	Identifier / raw metadata; Raw source field	Recorded in source data; intentionally not used as model input	Excluded from predictor matrices. Used only for temporal partitioning.

No.	Original feature name	English description / encoded level	Feature family & representation	Availability at intended prediction time	Leakage audit / final status
C2	KazaID	Crash identifier (used only for record integrity/split checks)	Identifier / raw metadata; Raw source field	Recorded in source data; intentionally not used as model input	Excluded from predictor matrices. Not used to construct a modeled outcome.
C3	KazaAyı	Raw crash month	Identifier / raw metadata; Raw source field	Recorded in source data; intentionally not used as model input	Excluded from predictor matrices. Not used to construct a modeled outcome.
C4	KazaGünü	Raw crash day	Identifier / raw metadata; Raw source field	Recorded in source data; intentionally not used as model input	Excluded from predictor matrices. Not used to construct a modeled outcome.
C5	KazaSaati	Raw crash hour	Identifier / raw metadata; Raw source field	Recorded in source data; intentionally not used as model input	Excluded from predictor matrices. Not used to construct a modeled outcome.
C6	KazaDakika	Raw crash minute	Identifier / raw metadata; Raw source field	Recorded in source data; intentionally not used as model input	Excluded from predictor matrices. Not used to construct a modeled outcome.
C7	KazaXKoordinat	Raw crash X coordinate	Identifier / raw metadata; Raw source field	Recorded in source data; intentionally not used as model input	Excluded from predictor matrices. Not used to construct a modeled outcome.
C8	KazaYKoordinat	Raw crash Y coordinate	Identifier / raw metadata; Raw source field	Recorded in source data; intentionally not used as model input	Excluded from predictor matrices. Not used to construct a modeled outcome.
C9	KazaYoladı	Raw road/street name	Identifier / raw metadata; Raw source field	Recorded in source data; intentionally not used as model input	Excluded from predictor matrices. Not used to construct a modeled outcome.
C10	TOPLAM_OLUYOLCU	Number of passenger fatalities	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Excluded from all predictor matrices to prevent direct or indirect target leakage.
C11	TOPLAM_OLUYAYA	Number of pedestrian fatalities	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Excluded from all predictor matrices to prevent direct or indirect target leakage.
C12	TOPLAM_OLUMLU	Fatal-crash indicator / recorded fatal outcome field	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Excluded from all predictor matrices to prevent direct or indirect target leakage.
C13	TOPLAM_YARALANMALI	Injury-crash indicator / recorded injury outcome field	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Excluded from all predictor matrices to prevent direct or indirect target leakage.
C14	TOPLAM_YARAYOLCU	Number of injured passengers	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Excluded from all predictor matrices to prevent direct or indirect target leakage.
C15	TOPLAM_YARAYAYA	Number of injured pedestrians	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Excluded from all predictor matrices to prevent direct or indirect target leakage.
C16	ToplamYoyaOlu	Total passenger/pedestrian fatalities	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Excluded from all predictor matrices to prevent direct or indirect target leakage.
C17	ToplamYoyaYarali	Total passenger/pedestrian injuries	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Excluded from all predictor matrices to prevent direct or indirect target leakage.
C18	SürücüYaralanmaŞiddeti	Driver injury-severity outcome-derived field	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Outcome-derived driver injury-severity field; explicitly excluded from the revised final predictor specification.
C19	TOPLAM_OLUSURUCU	Number of driver fatalities (source target for driver-fatality task)	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Primary source target field; excluded from all predictor matrices and used only to define/evaluate the corresponding modeled outcome.
C20	TOPLAM_YARASURUCU	Number of injured drivers (source target for driver-injury task)	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Primary source target field; excluded from all predictor matrices and used only to define/evaluate the corresponding modeled outcome.
C21	TOPLAM_YARALI	Total number of injured persons (source target for total-injury task)	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Primary source target field; excluded from all predictor matrices and used only to define/evaluate the corresponding modeled outcome.
C22	TOPLAM_OLU	Total number of fatalities (source target for total-fatality tasks)	Outcome / target-equivalent field; Recorded outcome field	Not admissible as predictor; used only for target construction/audit where applicable	Primary source target field; excluded from all predictor matrices and used only to define/evaluate the corresponding modeled outcome.

Interpretation note: This dictionary documents predictor availability and provenance for the revised manuscript. It does not imply that all retained variables are causal determinants; SHAP values are interpreted as model-attribution patterns within the defined post-crash record-based setting.